

APEXPLANET SOFTWARE PVT. LTD.

Cybersecurity & Ethical Hacking

Internship

TASK 1 — LAB SETUP REPORT

Foundation & Environment Setup

- **Objective:**

1. To build strong foundations in cybersecurity and set up a controlled cybersecurity lab environment using Kali Linux and Metasploitable 2.
2. The lab is configured on a private Host-Only network for safe testing, with Nmap used for network scanning and Wireshark used for packet capture.

• Lab Environment:

1. The cybersecurity lab was created using Oracle VirtualBox.
2. Kali Linux was configured as the attacker machine and Metasploitable 2 was configured as the intentionally vulnerable target machine.
3. Both virtual machines were connected through a private Host-Only network to provide an isolated environment for security testing.

Component	Configuration
Virtualization	Oracle VirtualBox
Attacker Machine	Kali Linux
Target Machine	Metasploitable 2
Network Type	Host-Only Adapter
Kali IP	192.168.56.101
Metasploitable IP	192.168.56.102

Virtual Machines:

Kali Linux: Kali Linux is a Linux-based penetration-testing and security assessment platform. It provides tools required for network discovery, vulnerability assessment, traffic analysis and other cybersecurity activities. In this lab, Kali Linux was used as the primary testing machine.

Metasploitable 2: Metasploitable 2 is a deliberately vulnerable Linux virtual machine designed for security testing and learning. It was configured as the target machine so that scanning and other controlled security-testing activities could be performed safely within the lab.

• Kali Linux Setup

1. Installation and Configuration

Kali Linux was deployed as the attacker machine using a pre-built VirtualBox image. Oracle VirtualBox was used to import and run the Kali Linux virtual machine. After successful deployment, the VM was started and its network interface was checked to verify connectivity with the isolated lab network.

2. Network Configuration

The Kali Linux virtual machine was connected to the Host-Only network used by the cybersecurity lab. The `ip addr` command was used to inspect the available network interfaces and confirm the assigned IP address.

Parameter	Value
Machine Role	Attacker
Operating System	Kali Linux
Network Interface	eth0
Network Type	Host-Only
IP Address	192.168.56.101
Subnet Mask	255.255.255.0

Home

File System

Trash

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff  
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0  
        valid_lft 405sec preferred_lft 405sec  
    inet6 fe80::c3bb:62f2:b117:feca/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
(kali@kali)-[~]  
$
```

• Metasploitable 2

1. Installation and Configuration

Metasploitable 2 was deployed as the intentionally vulnerable target machine for the cybersecurity lab. The existing Metasploitable virtual disk was imported into Oracle VirtualBox and configured as a virtual machine. The system was started successfully and the default msfadmin account was used to access the system.

2. Network Configuration

The Metasploitable 2 virtual machine was connected to the same Host-Only network as Kali Linux. This configuration allowed the attacker and target machines to communicate through the private lab network.

Parameter	Value
Machine Role	Vulnerable Target
Operating System	Metasploitable 2
Network Interface	eth0
Network Type	Host-Only
IP Address	192.168.56.102
Subnet Mask	255.255.255.0

To access official Ubuntu documentation, please visit:

<http://help.ubuntu.com/>

No mail.

msfadmin@metasploitable:~\$ ifconfig

```
eth0      Link encap:Ethernet  HWaddr 08:00:27:c3:a8:49
          inet addr:192.168.56.102  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fec3:a849/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:3 errors:0 dropped:0 overruns:0 frame:0
          TX packets:29 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1252 (1.2 KB)  TX bytes:3638 (3.5 KB)
          Base address:0xd020 Memory:f0200000-f0220000
```

```
lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:91 errors:0 dropped:0 overruns:0 frame:0
          TX packets:91 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19301 (18.8 KB)  TX bytes:19301 (18.8 KB)
```

msfadmin@metasploitable:~\$ _

• Network Connectivity Test

1. Objective

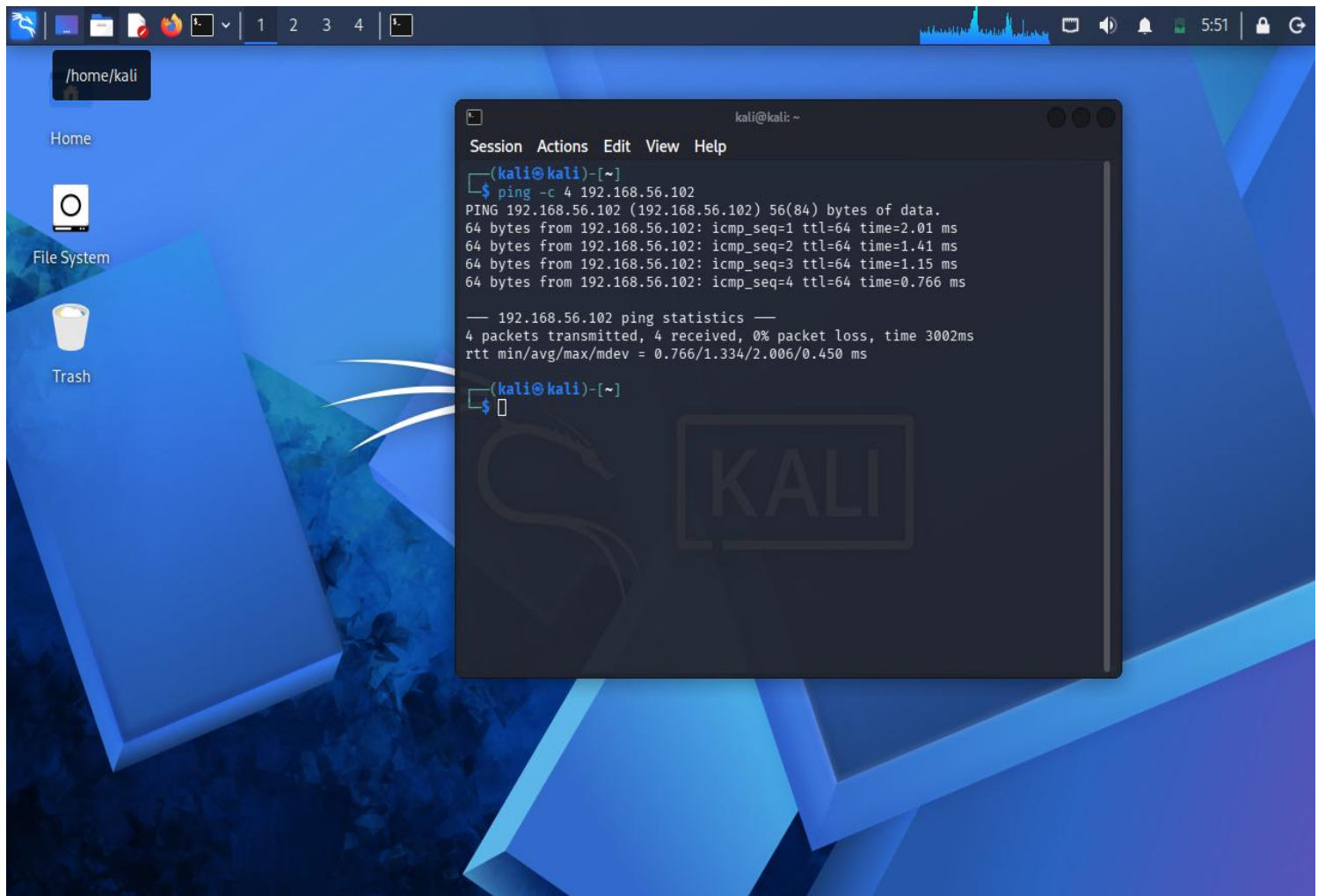
The connectivity test was performed to verify communication between the Kali Linux attacker machine and the Metasploitable 2 target machine over the private Host-Only network.

2. Ping Test

From Kali Linux, an ICMP ping test was performed against the Metasploitable 2 IP address using the following command:

The test generated four ICMP Echo Requests from Kali Linux to Metasploitable 2. All four packets received successful replies from the target machine.

Parameter	Result
Source	Kali Linux- 192.168.56.101
Destination	Metasploitable 2- 192.168.56.102
Packets Transmitted	4
Packets Received	4
Packet Loss	0%
Connectivity	Successful



```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ ping -c 4 192.168.56.102  
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.  
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=2.01 ms  
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=1.41 ms  
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=1.15 ms  
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=0.766 ms  
  
— 192.168.56.102 ping statistics —  
4 packets transmitted, 4 received, 0% packet loss, time 3002ms  
rtt min/avg/max/mdev = 0.766/1.334/2.006/0.450 ms  
(kali@kali)-[~]  
$
```

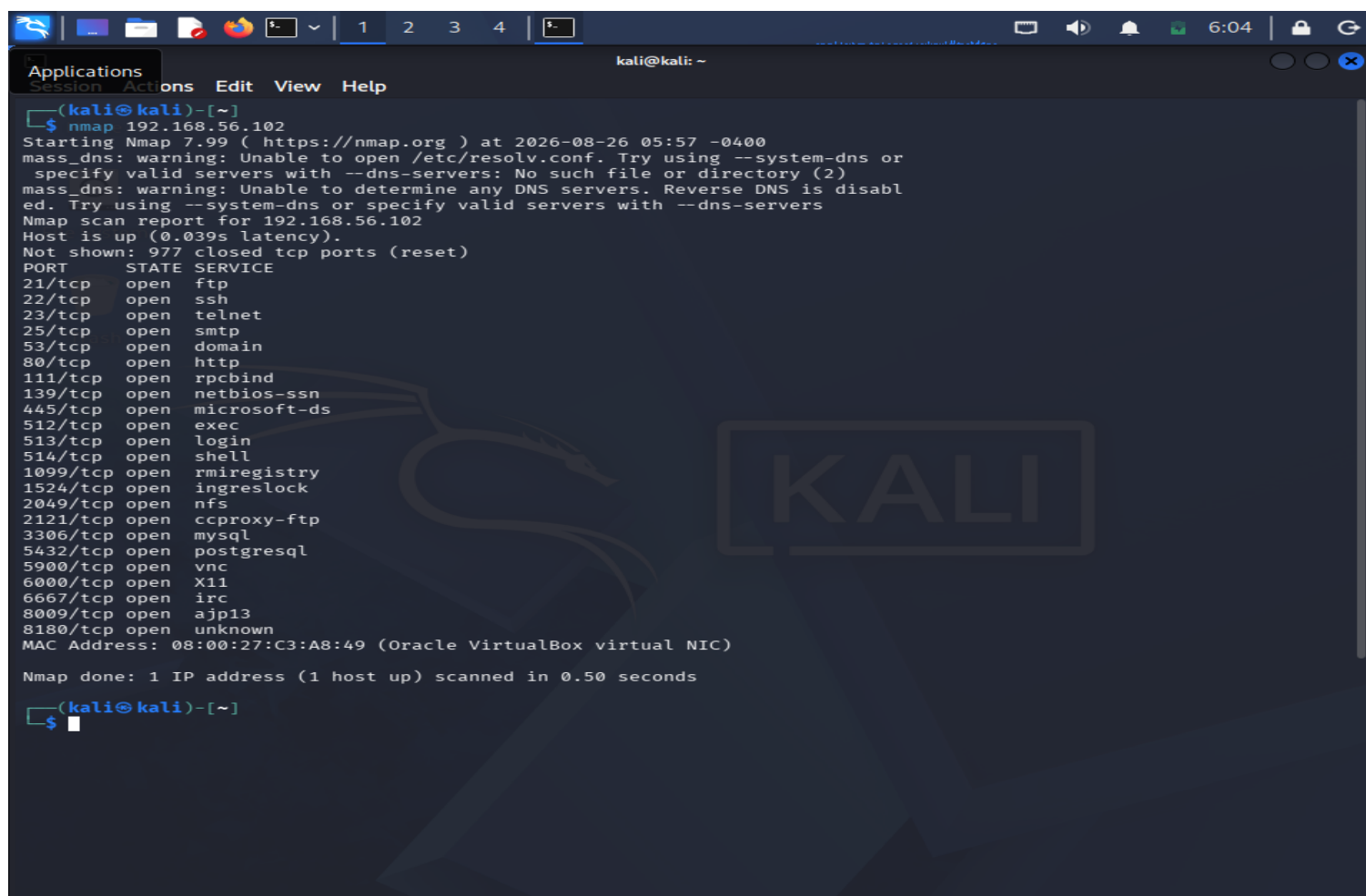
• Nmap Network Scan

1. Objective

Nmap was used to perform an initial network and service discovery scan of the Metasploitable 2 target. The purpose of the scan was to determine whether the target was reachable and identify the TCP services exposed by the target machine.

2. Scan Performed

The scan was executed from Kali Linux against the Metasploitable 2 IP address. Nmap confirmed that the target host was active and identified multiple open TCP ports and their associated services.



```
(kali@kali)-[~]
$ nmap 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-26 05:57 -0400
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or
specify valid servers with --dns-servers: No such file or directory (2)
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabl
ed. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.102
Host is up (0.039s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:C3:A8:49 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.50 seconds

(kali@kali)-[~]
$
```

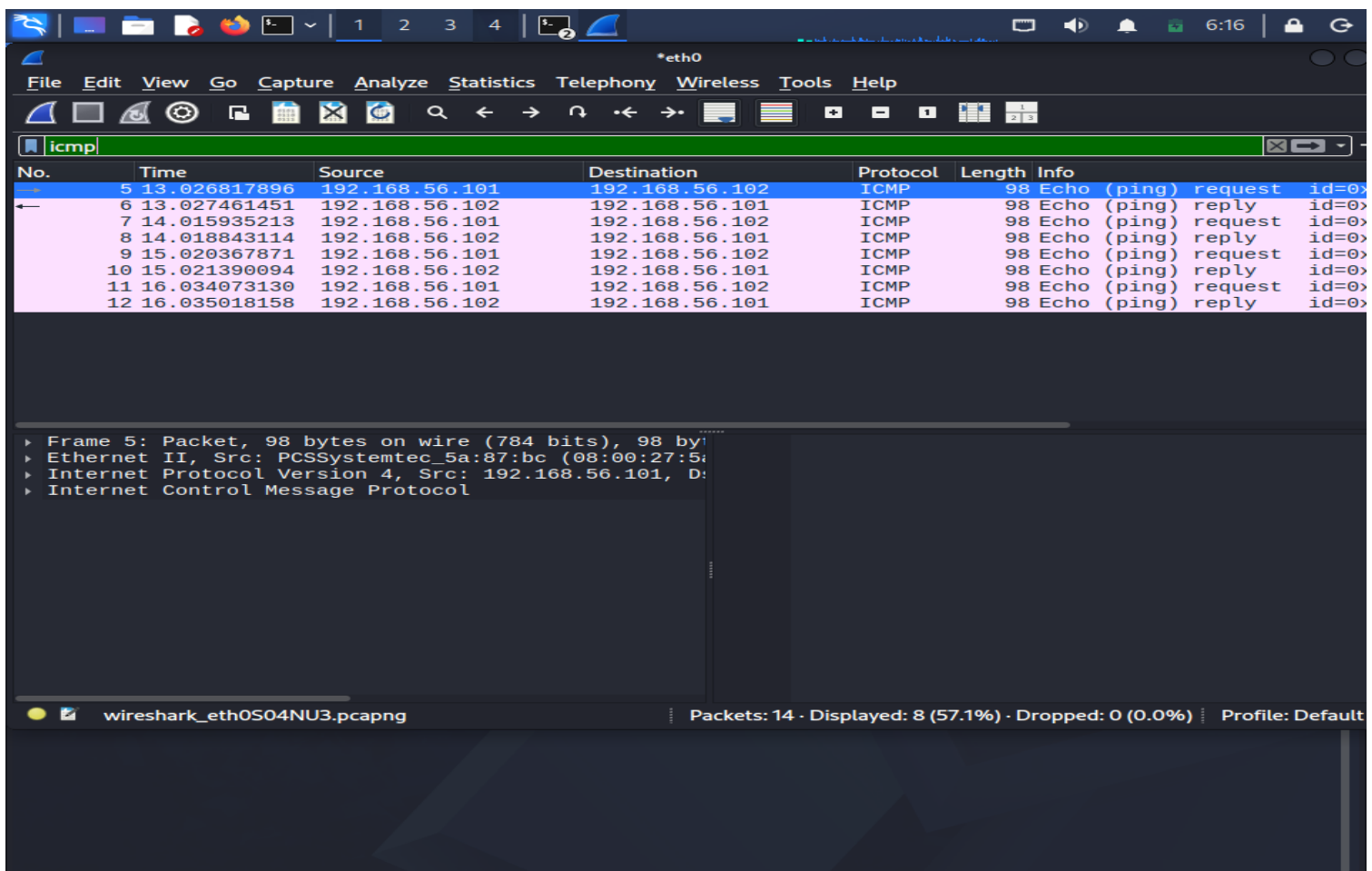
• Wireshark Packet Capture

1. Objective

Wireshark was used to capture and analyze network traffic exchanged between the Kali Linux attacker machine and the Metasploitable 2 target. The purpose of this test was to verify that network packets could be captured and examined within the controlled lab environment.

2. Capture Configuration

Wireshark was launched on Kali Linux and the eth0 interface was selected for packet capture. ICMP traffic was generated by performing a ping from Kali Linux to Metasploitable 2.



• Results and Observations

The cybersecurity lab environment was successfully configured using Oracle VirtualBox with Kali Linux and Metasploitable 2 connected through a private Host-Only network. Kali Linux was assigned 192.168.56.101, while Metasploitable 2 was assigned 192.168.56.102.

Connectivity testing was successful, with all four ICMP packets receiving replies and 0% packet loss. An initial Nmap scan successfully identified the target as active and revealed multiple open TCP services. Wireshark was also successfully used to capture and filter ICMP traffic exchanged between the two virtual machines.

Activity	Result
Kali Linux setup	Successfully completed
Metasploitable 2 setup	Successfully completed
Host-only network	Successfully configured
Nmap scan	Successfully completed
Wireshark packet capture	Successfully completed

• Conclusion

The Task 1 cybersecurity lab environment was successfully established and tested. The isolated network configuration provided a controlled environment for communication between the attacker and target machines. Basic network scanning with Nmap and packet analysis with Wireshark were successfully demonstrated.

